ROSHAN BHARATI

Systems & Security Engineer | C++, Python

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PROFILE

Systems & Security Engineer specializing in C++ systems, Windows internals, and network traffic analysis. Experienced in designing and shipping production software involving inline hooking, DLL injection, TLS interception, and Layer 7 traffic inspection, with a strong focus on performance, reliability, and maintainability. Comfortable taking ownership of complex systems, from architecture and implementation through testing, debugging, and documentation. Currently pursuing an M.Tech in Artificial Intelligence and researching application-agnostic Layer 7 content reconstruction for Data Loss Prevention.

PROFESSIONAL EXPERIENCE

Audio Bee · Web Scraping Consultant · Contract · Remote — *Mar 2026 – May 2026*

Built production extraction infrastructure for US health insurance plan data at scale, normalizing diverse carrier formats into the Ideon schema – a standardized interchange format for insurance APIs.

- Delivered **57 production scrapers in approximately four weeks** (~4 sites/day), handling volumes from ~10K to **2M+ records per site**, all normalized to a consistent downstream schema.
- Covered the full extraction spectrum: async REST/GraphQL API clients for structured endpoints, and Playwright-based browser automation for JS-heavy, **anti-bot-protected frontends** - requiring request fingerprint and session behaviour analysis to sustain throughput .
- Designed pipelines with clean API boundaries and fault-tolerant error handling for reliable delivery to downstream consumers.

GuardWare Australia · Systems Engineer · Kathmandu, Nepal — *Dec 2024 – Jan 2026*

Sole developer and full owner of three production C++ and Python systems shipped to customer environments — responsible for architecture, implementation, testing, and documentation.

Windows Endpoint Security Agent – Inline Hooking Framework · C++23, Microsoft Detours

- Designed and shipped a low-level Windows hooking framework using **Microsoft Detours** for real-time threat detection — deep work in memory layout, DLL injection lifecycle, and OS internals.
- Optimized hook execution using smart caching and lightweight call paths, keeping runtime overhead negligible in latency-sensitive endpoint processes. Deployed where stability and **sub-millisecond latency** were non-negotiable.
- Authored full technical documentation and a regression test suite; deployed across diverse customer machines with **zero critical defects post-release**.

TLS Interception Proxy — Architecture Rebuild · C++23, refactored from C++11

- Took sole ownership of a large, undocumented C++11 proxy with stability issues under live traffic; performed root-cause analysis and rebuilt its architecture as a modular C++23 framework.
- Rebuilt as a modular C++23 framework applying SOLID principles: reusable HTTP/JSON primitives, structured logging, and clear separation of concerns; **time to add new traffic-analysis coverage fell from days to hours.**
- Introduced unit and integration tests, converting an unmaintainable legacy codebase into a stable, extensible production system.

DLP Content Inspection – Archive Processing Module · Python

- Implemented a production backend module for secure archive handling (ZIP, 7z, TAR, GZ) in a **defense-grade data-discovery platform** — signature-based detection, password-protected archive support, stream-safe extraction.
- Delivered with structured logging, error classification, and full unit tests; integrated cleanly via a well-defined API boundary.

Darvis Studios · Backend Engineer · Part-time · Remote — *Jan 2024 – Jul 2024*

Video Processing Pipeline · Python, async

- Built and owned an async backend orchestrating an end-to-end AI video generation pipeline — **reduced runtime by ~40%** through async redesign and targeted optimizations.
- Designed for extensibility: new pipeline stages could be added with minimal coupling.

RESEARCH PROJECTS

DeepSeer – Layer 7 Content Reconstruction for DLP · C++, Python · Active Research · github.com/rosxnb/DeepSeer

- Investigating AI-based, **application-agnostic Layer 7 content reconstruction** for Data Loss Prevention – extracting semantic payload artifacts such as filenames, email headers and file content from encrypted web traffic **without service-specific protocol parsers**.
- Built a custom traffic interception and analysis platform (TLS, HTTP/1.1, HTTP/2) as the data collection and experimentation layer supporting model training, evaluation, and active-learning experiments.
- Exploring hybrid ML approaches for real-time content reconstruction and classification within **sub-100 ms inference budget**; targeting a research publication and open-source release.

PERSONAL PROJECTS

MaLA – Linear Algebra & Autograd Engine · C++23, SIMD, Metal · github/rosxnb/MaLA

- Built a from-scratch, **zero-third-party** linear algebra and reverse-mode automatic differentiation engine to understand the machinery underneath ML frameworks.
- Designed a clean, **layered architecture** (storage → tensor → kernels → ops → autograd) where lower layers have no awareness of autograd, enabling use as a standalone NumPy-like library without graph overhead.
- Implemented performance-critical kernels (GEMM, element-wise ops, reductions, im2col) using SIMD intrinsics and a thread pool, operating directly on plain pointers for testability and reuse across dtypes.

Metal GPU Tic-Tac-Toe · C++, MetalCPP, Objective-C++ · github/rosxnb/colored-tictactoe

- Built a real-time rendered Tic-Tac-Toe game for macOS using Apple's MetalCPP API, demonstrating cross-language integration between C++, Objective-C++, and Metal Shading Language.
- Implemented the full GPU pipeline setup, including shader programming for rendering and a custom math namespace for game logic, with keyboard-based navigation.
- Engineered a clean abstraction layer for input handling and game state management, showcasing modern C++ in a graphics programming context.

Lox Language Interpreter · C++17, CMake · github/rosxnb/lox-language

- Built a complete tree-walk interpreter from scratch for the Lox programming language, covering the full spec: lexer, recursive-descent parser, AST, and evaluator.
- Represented literals polymorphically using std::any, demonstrating thoughtful API design and a testable, extensible codebase.

OCR Backend Service · C++, Drogon, Tesseract, CMake · github/rosxnb/ocr_model

- Developed a REST API backend in C++ that integrates Tesseract OCR for image text extraction, using the Drogon web framework for asynchronous HTTP handling.
- Designed a clean POST endpoint to accept image uploads and return extracted text, demonstrating third-party library integration and robust API design.
- Built a complete CMake-based project with clear dependency management, run scripts, and a simple testable interface, reflecting an end-to-end development workflow.

Virtual Try-On System · Python, JavaScript · github/rosxnb/try-on-system

- Built a full-stack web application that renders a person's image fused with a chosen garment, providing a realistic virtual try-on experience.
- Integrated a custom-trained VITON (Virtual Try-On) model, covering the entire pipeline from pose keypoint detection and shape parsing to image warping and final render.
- Developed a complete system with a Python-based backend API (using FastAPI) and a JavaScript frontend, demonstrating end-to-end ownership from model inference to user interface.

TECHNICAL SKILLS

C++ Depth	C++11→23, STL, RAII, move semantics, templates, smart pointers
Windows Internal	DLL injection, inline hooking (Detours), memory layout, process instrumentation
Network Security	TLS interception, MITM proxy, HTTP/1.1/2, Layer 7 content reconstruction
GPU	Metal/MetalCPP, CUDA, OpenGL
ML	NLP/LLM, Computer Vision, Deep Learning

Tooling	CMake, Git, LLDB, Valgrind, Docker, Neovim, TMUX

EDUCATION

M.Tech in Artificial Intelligence 2025 – 2027 (Expected)
Kathmandu University

B.Sc. Computer Science and Information Technology 2019 – 2023
Tribhuvan University · Trinity International College

Cambridge GCE A-Level 2016 – 2018
Campion School, Lalitpur